



GE Medical Systems

gemedical.com

Technical Publication

Direction 2282382-100

Revision 01

**GE Medical Systems
LightSpeed QXi (1.3) Software Load
Procedure**

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- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

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- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
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ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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File a report with

- Call 1-800-548-3366 and use option 8.
http://data.supportcentral.ge.com/upload/15325/doc_324531.doc
- Fill out a report on <http://3.28.216.127/sctq/InstallFulfill/IFHome.htm>
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

End of Section

Revision History

Revision	Date	Reason for change
0	10/03/00	Initial Release of Publication for H1.3M3 (Software Version 9.X)
1	06/17/04	CQA 10210392 (p18, p41). CQA ctgce71617 (p18).

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Chapter 1

Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by its format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task. Please become familiar with the conventions used within this publication before proceeding.

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that are applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION



**WARNING
ROTATING
EQUIPMENT
BARE WIRES**

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.



**NOTICE
Equipment
Damage
Possible**

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles and shading are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

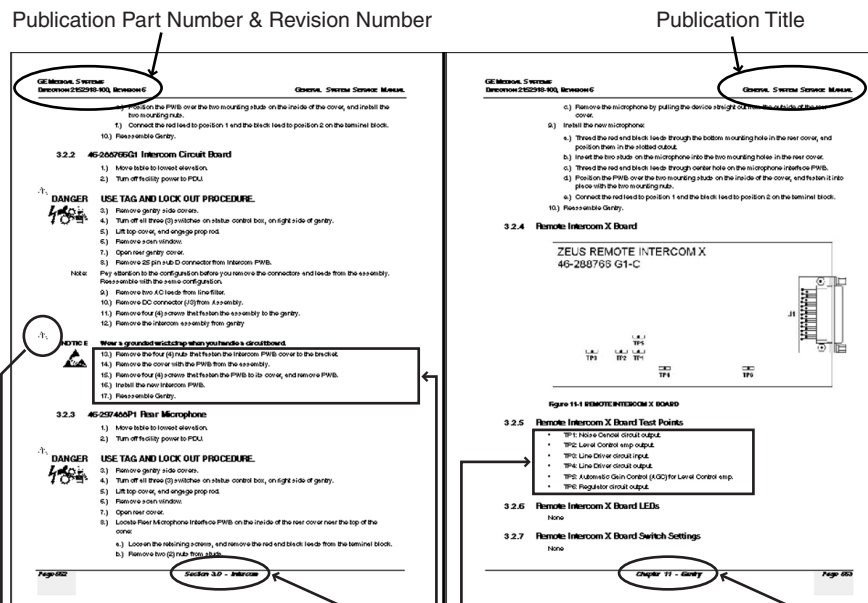
Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant.

2.2 Page Layout



The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by **Alphanumeric** (e.g. numbers) characters is information that must be followed in a **specific order**.

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by **symbols** (e.g. bullets) is information that has **no specific order**.

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicated direction. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example:
Fixed Output

This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application. It's the most commonly used style used to indicate filenames, paths and text.

Example:
Variable Output

This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_output>

Example:
Fixed Input

This paragraph denotes fixed input. It's typed input that will not vary from application to application. Fixed text the user is required to supply as input.

Example:
Variable Input

This paragraph denotes computer input that can vary from application to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example:
Hard Keys

A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example:
Soft Keys

Whereas the computer **MENU** button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

End of Section

Chapter 2

Load From Cold

Section 1.0 Introduction

1.1 Document Overview

Before you begin, please become familiar with the computer text and character representations used in the load from cold procedure. Please review the [Chapter 1 - "Publication Conventions"](#), section 2.3 on page 16 that cover computer text and screens.

The body of this publication consists of the following Sections:

- [Section 1.0 - "Introduction"](#): is the introduction to the Software "Load from Cold" Procedure.
- [Section 2.0 - "Load From Cold Procedure"](#): lists the step by step load from cold procedure for LightSpeed QX/i. Use and keep this document as a record of system information you might collect.
- [Appendix A - "Procedure For Creating Default Partitions"](#): deals with creating default partitions on new host computer (Octane) disk drives.
- [Appendix B - "Reconfiguration Procedure"](#): covers the procedure for performing Software Reconfiguration.
- [Appendix C - "Changing The System Time Zone"](#): deals with setting up Timezones on the OC and SBC (or OC only on an O2-less configuration console) using the Reconfiguration Procedure.
- [Appendix D - "Regenerating the Scan Database"](#): deals with regenerating the system's Scan Database (on the scan data disk) using the Reconfiguration Procedure.
- [Appendix E - "Installing Laser & DICOM Cameras"](#): covers installation, setup and advanced parameters. Also see the System Installation Manual, Chapter 12 for additional information and procedures regarding Networking.
- [Appendix F - "Error Messages And Troubleshooting"](#): gives additional help for errors not listed in the procedure directly, and potential solutions.

1.2 Software Load From Cold (LFC) Overview

LightSpeed CT Load From Cold (LFC) involves preparing the system disks, installing and configuring the system software, on both the OC and SBC computers (or on the OC computer only in an O2-less system).



NOTICE
Potential for
Data Loss

Make sure that all Images have been reconstructed and archived before performing this procedure. This procedure will re-initialize all system data disks, erasing all images and scan data.

Section 2.0

Load From Cold Procedure

2.1 Useful Commands

<code>mountMOD</code>	<i>(mount the MOD on the /MOD dir)</i>
<code>ls -l /MOD/service_mod_data/system_state</code>	<i>(or /MOD/serv*/sys*)</i>
<code>find /MOD -name '*abc*' -print</code>	<i>(locates any file with 'abc' in it)</i>
<code>cd; umountMOD</code>	<i>(must cd out of MOD dirs to unmount it)</i>
<code>more /usr/g/config/SysState.cfg</code>	<i>(view locations of System State files)</i>
<code>su -; (password); man 1 chmod chown</code>	<i>(access man pages as root)</i>
<code>rsh sbc</code>	<i>(access the SBC)</i>

2.2 Install Software - Contained on (4) CD-ROMS

- 1.) LightSpeed QXi IRIX Operating System CD-ROM Disk #1
- 2.) LightSpeed QXi IRIX Operating System CD-ROM Disk #2
- 3.) LightSpeed QXi IRIX Operating System CD-ROM Disk #3
- 4.) LightSpeed QXi Applications Software CD-ROM

2.3 Software Revision Check

The software revision can be checked using the showprods command.

- 1.) Open a UNIX shell.
- 2.) Type:

```
showprods | grep -i "QXi Apps" ENTER
```

- 3.) If you see:

```
<6.8.2_2.7.3n2> STOP! System is at revision 1.1. Perform FMI 25263.  
<7.2.1_2.8.2F_HE1.2M3>, <8.12.2_2.8.2G_H1.2M4>, or <10.4.1_X.X.XX_H1.3M4>  
STOP! System is at revision 1.2 or a preliminary version of 1.3 software. Perform FMI 25269.
```

2.4 Image Data Archive

The following procedures will destroy all image data on the system. Verify that the customer has reconstructed and archived all images before proceeding.

2.5 System Operation Verification

Setup and perform at least one axial scan to verify that the system operated properly before beginning the upgrade.

2.6 Remove DASM From SCSI Chain

During the load of applications software, the SCSI bus is being heavily utilized. To minimize the chance of problems occurring as a result of this heavy use, it is recommended that the DASM be

removed from the SCSI bus chain, before continuing with the software load. Once the LFC has been performed, the DASM can be reinstalled.

To remove the DASM from the SCSI bus chain:

- 1.) At the SCSI Expansion box, disconnect the cable coming from the DASM.
- 2.) At the DASM, remove the SCSI cable that comes from the Octane and connect it to the SCSI Expansion box connection opened in step one.

2.7 Testing CD-ROM Drive Operation

Since the CD-ROM Drive is seldom used except for loading system software, it is advisable at this time to perform a quick test on the ability of the CD-ROM drive to read data from a CD before you begin the LFC procedure. This test is accomplished by inserting and mounting one of the OS CD-ROM disks, and performing a checksum (read check) on the files in one of the directories on the CD. If any errors occur while the system is reading this CD-ROM, you must stop to investigate and correct the CD problem (for example the SCSI Bus, CD-ROM Drive, etc. problems).

- 1.) Open a UNIX shell.
- 2.) Type: `tail -f /usr/adm/SYSLOG` ENTER
- 3.) Open a second UNIX shell.
- 4.) Insert the LightSpeed QX/i IRIX OS CD-ROM Disk #1 into the CD-ROM drive, and type:
`mountCD /mnt` ENTER
`cd /mnt/dist` ENTER
`sum *` ENTER
- 5.) While the system is reading the CD-ROM and calculating checksums, you need to monitor the SYSLOG in the first UNIX shell window for any error messages that may occur that are related to SCSI or CD-ROM problems (timeouts, read errors, SCSI Bus resets, etc.).
- 6.) If checks are successful (no errors produced), you can continue with the LFC procedure. Continue to [Section 2.0](#).
- 7.) If you encounter any CD-ROM or SCSI related errors during this check, stop and troubleshoot/repair the problem before attempting to perform an LFC.
- 8.) Terminate the 'tail' command by typing ^c in the first UNIX shell window where the 'tail' command is running.
- 9.) Unmount the CD-ROM by typing the following commands in the second UNIX shell window:
`cd /` ENTER
`unmountCD` ENTER

2.8 Save System State Data

Before beginning the software install, you will want to save the system configuration information, characterization, calibration, protocols, etc. to a System State MOD.

- 1.) Bring the system up if it is not already up.
- 2.) Insert the System State MOD.
- 3.) Click on the SERVICE DESKTOP.
- 4.) Click on the PROACTIVE/PREVENTIVE/PLANNED MAINTENANCE icon.
- 5.) Click on SYSTEM STATE.
- 6.) Click on ALL. This will highlight Cals, Characterization, Reconfig Info, etc.
- 7.) Click on SAVE.

The save will take a few minutes. Review the output for errors or missing files; the scroll bar on the right works only when the tool isn't busy performing some task, it may take a little while. If you see any missing files or failures, then refer to the note below.

- 8.) Click on DISMISS.

If System State reported no problems, then you may proceed to [section 2.9](#)

Note: **IF YOU SHOULD GET SAVE/RESTORE SYSTEM STATE ERRORS:**The Save/Restore System State process reports status on each file specification listed in the `/usr/g/config/SysState.cfg` file. The status may be:

- Save (or Restore) of {filename} succeeded
- Save (or Restore) of {filename} skipped. File not found
- FAILED

A status of `FAILED`, indicates potential problems, `skipped` means the file not found was not required. Please review the following for each file you see an error reported.

Some files simply may not exist on the system. If certain tools were never run then certain files may never get created. You may view `/usr/g/config/SysState.cfg` and look for a required field. `req=Y` means the file in question is required. You may also wish to check the system and MOD to verify whether it was actually missing or if there were permission or owner problems with the file.

2.9 Record KEY System Information

Keep a record of system information in a safe place. Although system information is saved on the System State MOD, an additional record should be kept elsewhere at your site, in case the State MOD is lost or damaged. This step may be optional, if this information is already recorded elsewhere.

- 1.) Record the most important Reconfig INFO for the scanner. Use the following procedure to help gather key system information from the INFO file and other system files.

- a.) Open or go to an UNIX shell on the OC.

- b.) View the following files to find the information for the chart in [Table 2-1](#). Type:

`more /usr/g/config/INFO` ENTER to get all data except Exam Number and Camera information. No HIS/RIS information will be present if the customer does not have this option.

`more /usr/g/bin/scanRx.db` ENTER to get the next Exam Number
(Add 1 to the number to record the "next" exam number.)

For each of the fields below, look at the `more` file output to find the associated `setenv` value and record the value in [Table 2-1](#).

Note: **IP Addresses and Net/Broadcast Masks have the following format: `www.xxx.yyy.zzz`**

System Network Configuration			
	Field Name:	"setenv" Name:	Field Value:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number**	Use procedure above to determine Exam Number.	
	DASM Type	DASMTYPE	See Appendix E
	Camera Type	CAMERATYPE	See Appendix E
Network Settings:	Host Name	GATEWAY_HOSTNAME	

Table 2-1 Re-configuration Information

System Network Configuration			
	Gateway IP Address	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Default Gateway	GATEWAY_DEFAULT	
	Suite Name	SUITEID	
Optional - Network Printer	Printer Hostname	PRINTER_HOSTNAME	
	Printer IP Address	PRINTER_IPADDRESS	
Optional - HIS/RIS Systems*	CT AE Title	HRCTTITLE	
	Port Number	HRPORTNUM	
	H/R Server AE Title	HRAETITLE	
	IP Address	HRIPADDR	

Table 2-1 Re-configuration Information

Note: * HIS/RIS information will not be in the INFO file if the customer does not have the HIS/RIS option.
 ** Do not record the Exam Number from the data contained in the INFO file, because the value will be from the last time reconfig was performed. See the procedure on the previous page to determine the current Exam Number.

- c.) Go to [Appendix E - "Installing Laser & DICOM Cameras"](#), Table 2-5 on page 65 or Table 2-6 on page 66 to record camera information in the appropriate table provided.
- 2.) Now record the Networking Application (Image transfer) Configuration information, see [Table 2-2](#). On the ImageWorks Desktop, in the Browser under Network, for each host configured for networking do the following and record the information in the table provided:
 - a.) Click on IMAGE WORKS
 - b.) Click on NETWORK
 For each host configured for networking, perform the following steps, and record the information in Table 2-2 on page 23
 - c.) Click on SELECT REMOTE HOST.
 - d.) Click on the HOSTNAME.
 - e.) Click on UPDATE.
 - f.) Write the information in the table provided on the next page.
 - g.) Click on CANCEL to exit Select Remote Host.

If you find this process tedious, you can look at the file by typing:

more /usr/g/ctuser/Prefs/SdCRHosts

on the system to get this information. The 1 in the file means Advantage Net, the 2 means DICOM. If you see the pound sign (#), it's used as a separator and means nothing.

TYPICAL OUTPUT WOULD APPEAR AS THE FOLLOWING:

A	B	C	D	E	F	G	H	I	J
3.7.52.135	engbay04	#	2	4006	engbay04	No	comment	0	1 1 0
3.7.52.151	engbay26	#	2	4006	engbay26	No	comment	0	1 1 0

```
3.7.52.121 engbay16 # 1 4005 engbay16 No comment 0 1 0 1
```

The fields in the above file relate to the Network GUI settings.

A - IP Address

B - Hostname

C - # is a space, 1 is Advantage Net Protocol, 2 is DICOM Protocol

D - Port Number

E - AE Title

F - Comment Fields

G - Archive Node - 1 = Yes, 0 = No

H - Send Images - 1 = Yes, 0 = No

I - Query/Retrieve - 1 = Yes, 0 = No

J - Custom Search - 1 = Yes, 0 = No

- 3.) Check the memory to ensure it is recognized.

Open a UNIX shell and type: **hinv**

The output for the OC memory should be at least 512MB. If it is not, check the Host computer to make sure the memory is installed. If it is not installed, order the additional memory.

Note: The System must have at least 512MB Memory for proper operation. System performance will be seriously impaired if less than 512MB is present.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION							**	**	**	**	**
Host name	Network address	Network protocol	Port number	AE Title	Archive Node	Send Images	Query/ Retrieve	Custom Search	Comments		

Table 2-2 Network Application Configuration

 **NOTICE** Make sure that all Images have been reconstructed and archived before performing this procedure. This procedure will re-initialize all system data disks, erasing all images and scan data.

Potential for Data Loss

2.10 Shutdown the System

Click on the SHUTDOWN button (shown in [Figure 2-1](#)). An attention box will pop up indicating "Shutdown the System?"; press OK to shutdown Applications. This will shutdown both QX/i Applications and UNIX, and put you at the boot prompt.



Figure 2-1 Shutdown Icon

The following message should now come up:

Okay to power off the system now. Press any key to restart.

2.11 Boot Standalone IRIX Disk Partitioning Utility

- 1.) Power off the Console and wait for 15-20 seconds to reset everything properly.

Note: The power-off reboot is necessary to cleanup the swap space of the SBC before the OS load (on a console that contains an O2 computer), so that we will not get "not enough space to load miniroot" error during the SBC load.

- 2.) Please read carefully before proceeding.

You may notice a message about power up diagnostics running. Then a window will come up about starting the system. There will be a small button in the right hand corner named STOP FOR MAINTENANCE for 10 seconds after the console is powered ON. Using the mouse, quickly press this button when it appears.



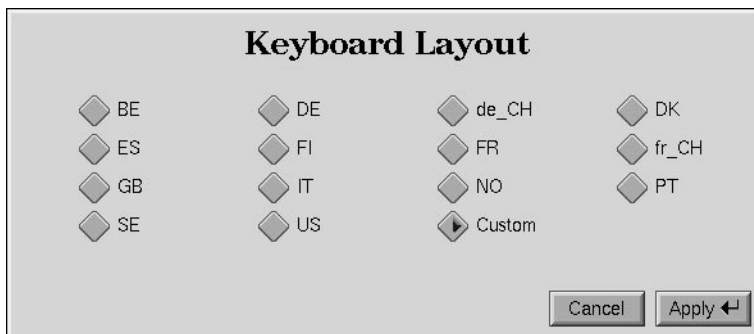
Now, power on the console and select STOP FOR MAINTENANCE.

- 3.) A screen will pop up with several icons displayed. The selections on the System Maintenance Menu are:
 - a.) Start System
 - b.) Install System Software
 - c.) Run Diagnostics
 - d.) Recover System
 - e.) Enter Command Mode
 - f.) Select Keyboard Layout
- 4.) Insert the LightSpeed QX/i IRIX Operating System CD-ROM Disk #1 into the CD-ROM drive.

2.12 Procedure for Non-US-English Keyboards

If you have a non-US keyboard, you should follow the procedure that follows. Else you may proceed to section 2.13 on page 25.

- 1.) Using the mouse, click on SELECT KEYBOARD LAYOUT



- 2.) Select the appropriate keyboard type. Supported keyboards are:
 - English - US (default)
 - German - DE
 - Swedish - SE
 - French - FR
- 3.) Select APPLY to reconfigure.

2.13 Install System Software (IRIX 6.5 OS) on Operator Console (OC)

This procedure assumes you have already created default partitions. You need to follow the procedures for creating default partitions in [Appendix A - "Procedure For Creating Default Partitions"](#) *ONLY IF*:

The disk is new, the disk has just been formatted, or if0 the disk has been corrupted.

Otherwise continue on to [step 1](#).

- 1.) Click on INSTALL SYSTEM SOFTWARE.
Make sure that the LightSpeed QX/i IRIX Operating System CD-ROM Disk #1 is in the CD-ROM drive and *the CD-ROM drive is not busy before proceeding to the next step*.
- 2.) Click on LOCAL CD ROM. (It may be LOCAL SCSI CDROM DRIVE 6 ON CONTROLLER 1)
- 3.) Click on INSTALL to continue installation.
A message appears: `Insert the installation CD-ROM now`
- 4.) Click on CONTINUE.
The following message will appear: `Obtaining installation tools`
Another message then appears: `Copying installation tools to disk`
The screen will go blank and another message will appear: `Creating miniroot devices, please wait ...`

Comment:

If you do not get one of the following 4 error messages, proceed to [step 5](#).

- If the message `Unable to continue; press <Enter> to return to the menu` appears on screen, do the following:
 - a.) Eject the CD from the CD-ROM drive, and re-insert the CD into the drive.
 - b.) Click on the RETURN button to continue.

- c.) Click on the STOP FOR MAINTENANCE button.
- d.) Return to [step 1](#) of this section.
- If the message `make new file system on /dev/dsk/dks0d1s0[yes/no/sh/help];` appears on the screen, do the following:
 - a.) Type: **yes** and press ENTER.
 - b.) When message `Are you sure ?[y/n] (n)` appears, type: **yes** and press ENTER.
 - c.) When message `Blocksize Of filesystem 512 or 4096 bytes ?` appears, type: **512** and press ENTER.
- If the message `Unable to mount partition /dev/dsk/dks0d1s0` appears on screen do the following:
 - a.) Press ENTER to go to a `cs` shell.
 - b.) Press ENTER and in the shell, type: **mkfs /dev/dsk/dks0d1s0**
 - c.) Type: **exit** ENTER and proceed to [step 5](#) on page 26.
- If the disk has never been partitioned, an error condition is reported when starting to partition the disk: `swap partition (1) is not a valid swap area .`
Refer to [Appendix A - "Procedure For Creating Default Partitions"](#) before continuing.
- See [Appendix F - "Error Messages And Troubleshooting"](#), page 69 for additional messages.

5.) The install menu should appear: Go to the next step if/when the menu appears.

Example:
Screen Output

```
Default distribution to install from: /CDROM/dist
For help on inst commands, type help overview.
Inst Main Menu
 1. from [source]      Specify location of ...
 2. open [source]     Specify additional ...
 3. close [source]    Close distributions
 4. list [keywords] [names] Display information ...
 5. go                Perform software ...
 6. install[keywords] [names] Select subsystems ...
 7. remove[keywords] [names] Select subsystems ...
 8. keep [keywords] [names] Do not install or ...
 9. step [keywords] [names] Interactive mode ...
10. conflicts [choice ...] List or resolve ...
11. help [topic]      Get help in general ...
12. view ...          Go to the view ...
13. admin ...         Go to the Admin ...
14. quit              Terminate software ...
inst>
```

- 6.) Type: **sh** and press ENTER.
Ignore the message: `Can't open /etc/sys_id`
- a.) At the prompt: `#` type: **mount /CDROM** and press ENTER.
 - b.) At the prompt: `#` type: **/CDROM/ct_prep** and press ENTER.
Messages appear: (Several pages of scrolling text appear)
`Starting automatic disk partition script`
`Ensuring the unnecessary partitions are unmounted`
`Copying Miniroot Image to the new partition location`

It may take a few minutes...

Note: Please ignore message: UX:dd: ERROR:Read error: No space left on device

Copying contents of the first OS Disk (Disk #1) to working directory
...

Comment: After a wait of about 5 minutes, you will see the message:]

Eject the first OS Disk #1

Please insert the second OS Disk (Disk #2), then press ENTER.

- c.) Replace the IRIX CD-ROM Disk #1 in the drive with the IRIX CD-ROM Disk #2 and press ENTER.

Comment: You will get a number of messages:

Copying Copying packages from the second OS Disk #2 to working directory...

Comment: After approx. 5 minutes, you will see the message:

Eject the second OS Disk #2

Please insert the third OS Disk (Disk #3), then press ENTER

- d.) Replace the IRIX CD-ROM Disk #2 in the drive with the IRIX CD-ROM Disk #3 and press ENTER

Comment: A message appears:

Copying packages from the third OS Disk #3 to working directory...

The packages from the last OS disk will be copied and the installation process will continue. This process will take approximately 30-40 minutes and no user input is required.

Note: During this time, files will be uncompressed, installations and removals will take place, but no errors should be generated. If any errors do occur during this process, refer to [Appendix F - "Error Messages And Troubleshooting"](#), page 69.

After the OS installation is complete the user will be instructed to type in the following commands to complete the IRIX OS(OC) install.

***** Done installing the OC System *****

- e.) Type: **'exit'** to exit back into the original Inst session
f.) Type: **'exit'** again at the Inst> prompt, restart the system.
g.) Type: **'yes'** when you are prompted to restart the system...

The system monitors will go blank and restart the OC Operating system, and automatically log you into the system as the root user on the SGI standalone workstation desktop.

Note: Ignore the following messages during OC bootup:

Failed to configure ef1...

exportfs: /CDROM: No such file or directory

2.14 Installing System Software (IRIX 6.5 OS) on the SBC (for Systems with an O2 Computer)

Note: This section ONLY applies to consoles with an O2 Computer installed. To determine if this console has an O2 Computer installed, either open the front cover of the console and look inside to see if an O2 is present, or look at the Console Model Number on the Console Rating Plate. Console Model #2180551 contains an O2 Computer; perform the procedure in this section. Console Model #2180551-2 does not contain an O2; skip this procedure and go to section 2.15 on page 30.

PREPARATION

Make sure that the LightSpeed QX/i IRIX OS CD-ROM Disk #3 is in the CD-ROM drive.

If you need to check, open a UNIX shell and eject the CD from the CD-ROM drive by and typing:
`eject 1 6 ENTER`

PROCEDURE

If errors occur, see [Appendix F - "Error Messages And Troubleshooting"](#), page 69 for possible recovery.

Comment: To view detailed debug information from the console window, Type:
`tail -f /tmp/autoSBC.debug`

- 1.) Under the TOOLCHEST, select DESKTOP, then OPEN UNIX SHELL.
- 2.) Within the UNIX shell, Type: `/usr/bin/autoSBC ENTER`

Comment: The following message appears:

```
*** Performing Load from Cold on the SBC ***
Connecting to the SBC ...
Connection established.
Please Power Cycle the SBC, I will wait ....
```

Note: If the program is waiting at this prompt for longer than 15 seconds, you are required to power cycle the SBC. Remove the front cover from the console and press the power button on the O2 computer (the power button is located to the left of the red/green LED's on the front of the O2 computer). After the O2 power cycle, the system will continue to load.

Comment: Messages appear:

```
Formatting the SBC Disk ...
SBC Disk Formatting Complete.
Setting up the network install from the OC.
Copying installation program to disk ...
Creating miniroot devices, please wait ...
Remaking (mkfs) file system on: /dev/dsk/dks0d1s0
Setting up network communications to the OC ...
Network setup complete.
```

Comment: *Messages appear:*

Note: If you see the following message:
 Copying Miniroot Image to the new partition location
 It may take a few minutes ...
 The system will appear not to be responding
 This may take approximately 5 - 10 minutes, please be patient.
 - - - Installation messages continue.

 Getting ready to install the SGI OS on the SBC.
 We are done loading the SBC ... Restarting the OS.
 Rebooting the SBC, One moment please ...

Note: The complete load takes approximately 40 minutes. After it's complete, you will be instructed to install the CT Applications. The instructions are listed in the following section.

2.15 Install QX/i Software on the OC and SBC (or on the OC Only, on an O2-less System)

Note: If you have problems, see [Appendix F - "Error Messages And Troubleshooting"](#), beginning on page 69 for possible error messages and their recovery.

- 1.) On the OC, open an "UNIX Shell" window by using the "Desktop" pull down menu.
- 2.) To ensure that the IRIX media daemon does not interfere with the "install.qx" script, stop it by typing:
`su - ENTER` And enter the root password to login as root.
`/etc/init.d/mediad stop ENTER`
- 3.) Eject the IRIX Operating System CD-ROM Disk #3 by typing: `eject 1 6 ENTER` in the OC UNIX shell.
- 4.) Insert the LightSpeed QX/i Application Software CD-ROM into the CD-ROM drive.
- 5.) Make sure that the System State MOD is in the MOD drive and is not busy.
If you need to check the MOD, to eject the MOD cartridge, Type: `eject 1 3 ENTER`
- 6.) Wait until the CD-ROM drive is not busy. Then type:
`cd /CDROM ENTER`
`install.qx ENTER`

Comment:

- 1.) If the message `/mnt/install - no such file or directory` appears, or if the CD-ROM fails to mount, proceed as follows:
 - a.) Type: `umount /mnt` and press ENTER
 - b.) Eject and then re-insert the CD-ROM media into the CD-ROM drive
 - c.) Return to [step 6](#) of this section.
- 2.) If your system contains a TEAC CD-ROM drive, when you execute the command `install.qx` for the first time, it may fail and display a message that says:
"mount_iso9660: Note: unknown CD-ROM type may not function correctly, Inquiry12 type = <TEAC CD-ROM CD-532S 1.0A>".
If this occurs, cancel, and then execute the command again, and it will function properly.

- 7.) Wait for the window shown in [Figure 2-2](#) to appear before proceeding to the next step.

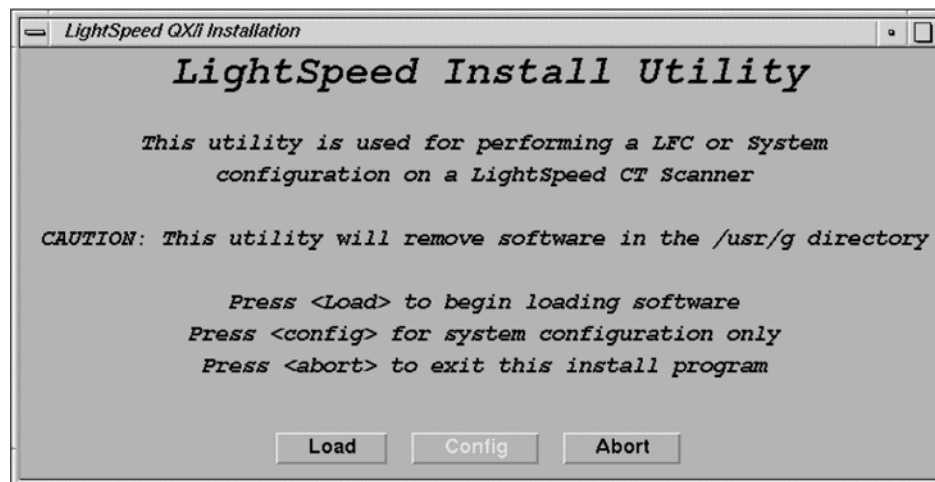


Figure 2-2 Install Utility Window

In the Install Utility window, click on LOAD. A pop up message will query if you wish to load the INFO file from the System State MOD; if you have a System State MOD, select YES -- another pop-up message will appear and you should ensure the System State MOD is in the drive and select OK.

- 8.) Configure System Settings: In the following steps, you will now go through each of the Reconfig screens and verify the information. You may make updates as needed. If you did not have a System State MOD, you will need to enter the appropriate information (refer to the data you recorded in section 2.9 on page 20).

Figure 2-3 System Settings Screen

- a.) Change the `Hospital Name` to the site's preferred name.
- b.) Enter the `Service ID` issued by the service organization.

Note: Use the scrollbar at the bottom of the timezone selection list to view the entire description of the timezone you are about to select, to ensure that you are selecting the correct timezone for your location.
If the time zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect.

- c.) Verify and/or change `Next Patient Exam #` (it should be (1) larger than the "last exam number" recorded earlier with KEY system information in section 2.9 on page 20).

Note: The `Next MOD#` and `Next Diagnostic Exam #` screens are currently not implemented.

- d.) Select PREFERENCES to proceed to the screen in the next step.

9.) Configure Preferences Setup Screen:

Figure 2-4 Preferences Setup Screen

Note: The Doctor's Title and Units for patient weight options are not currently implemented.

- a.) Select the Language (ENGLISH, FRENCH, GERMAN, ITALIAN) to display on the system.
- b.) Select Preferred FastCal KV - select the kV to be calibrated during FastCal. (Default selections are 120 and 140)
- c.) Select the desired Date Format and the desired Time Format.
- d.) Default Archive Device is DICOM.
- e.) Select HARDWARE to proceed to the screen in the next step.

Comment:

These kVs should include all kVs which the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs

10.) Configure Hardware Settings Screen:

Figure 2-5 Hardware Settings Screen

- a.) Select the DAS Type for this system. At present time, only S DAS is supported.
- b.) Select the Tube Type for the system. At present time, there is only one Tube Type.
- c.) Select the PDU Type for the system. At present time, only CT_COMPACT is supported.
- d.) Select the desired Collimator/Filter Type. At present time, there is only one collimator selection.
- e.) Verify the presence or absence of an O2 computer in your console, and select YES or NO as appropriate.

Note: To verify the presence or absence of an O2 Computer in this console, either open the console and look, or verify the model number on the console Rating Plate. Console Model #2180551 contains an O2 Computer. Console Model #2180551-2 does not.

- f.) Always select H1 GANTRY for Gantry Type.
- g.) If you have a Network Image Printer, do not select it at this time. You will need to run Reconfig (See [Appendix B on page 45](#)) after Applications load has completed, to set it up..

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e.: a computer with a printer attached); a Network Image Printer (which is supported) is a printer directly attached to the network.

- h.) Camera setup:
Comment: Cameras will be installed later in the load through the Utilities menu in the Service Desktop.
- i.) Select NETWORK to proceed to the screen in the next step.

11.) Configure the Network Settings screen.

LightSpeed QX/i System Configuration

System Default Preferences Default Hardware Default Network Default Accept Quit

Network Settings

Gateway Parameters

Suite Name CT01

Host Name ct01

IP Address 192.9.101.1

Net Mask 255.255.255.0

Broadcast Address 192.9.101.0

Default Gateway

☐ Advanced Options

This is also the AE Title of the scanner

Figure 2-6 Network Settings Screen

Note: Refer to the data you recorded in [section 2.9](#) to verify these entries.

a.) Enter the Suite Name.

The Suite Name must start with a letter, followed by 3 alphanumeric characters (i.e.: it MUST be 4 characters long). The name of the OC interface will be <Suite Name>_oc and SBC interface will be <Suite Name>_sbc (for systems that contain an O2 Computer).

b.) Enter the Host Name.

The Host Name identifies the hostname and AE Title of the scanner. The Host Name CANNOT be <Suite Name>_oc or <SUITE NAME>_OC .

c.) Enter the Host System's IP Address.

If possible consult the site's network administrator before configuring the interface to the site's network. The site administrator should provide the System's Host Name, IP Address, Net Mask, Broadcast Address, and Default Gateway (if a Default Gateway is used at the site).

d.) If you changed the Host System's IP address from the default then the Broadcast Address also needs to be updated. The Broadcast Address should be the Gateway IP address with the bits of the host id portion of the address either set to 1s or 0s depending on the configuration of the network. The standard default is 1's but older SunOS machines used 0's.

Comment:

If the Host System's IP address is 192.100.9.17 the broadcast address should be 192.100.9.255 if the network is configured to use 1s to specify the broadcast address. If the network contains genesis based scanners or other SunOS 3.5 or 4.1 machines the broadcast address should be 192.100.9.0.

e.) Enter the Default Gateway IP Address (optional).

If there is a Default Gateway for the site's network, enter the IP Address of the gateway. Otherwise, leave the field blank.

- f.) Check/select ADVANCED OPTIONS. The Advanced networking options provide support for NIS.

Do Not turn on NIS unless requested by the site's network administrator. Turning on NIS by selecting the Use NIS? option allows the scanner to use the site's NIS (a.k.a. Yellow Pages) database. The Domain Name must be provided by the site's network administrator.

Comment:

- * To turn on NIS:
 - 1.) Highlight ADVANCED OPTIONS.
 - 2.) Highlight USE NIS?.
 - 3.) Enter Domain Name
- * To Turn off NIS:
 - 1.) Highlight ADVANCED OPTIONS.
 - 2.) De-select USE NIS.

You have now readied the system to initiate installation.

- 12.) Select ACCEPT to utilize the configuration and continue installation.
- 13.) The system will load the CT software, OS patches, kernel changes and configure the system on both OC and SBC (or OC Only on an O2-less System).
- This loading process will take about 15 minutes. While the load is going on, the results will be displayed in a shell window (which closes when the load is complete). All the window output is logged to a file named `/var/adm/install.log.YYYYMMDDWWHHMMSS`. (Where `YYYYMMDDWWHHMMSS` is the Date/Time that the process was started.)
- 14.) When the loading process and configuration changes are complete, the system will prompt you to reboot. Click on YES.
- 15.) The system will automatically login in as `ctuser` after the reboot. Select OK on the `AutoStart Disabled` popup message.

2.16 Set Date/Time on SBC & OC (or on OC Only if an O2-less System)

You must always set date and time on the SBC and the OC (OC Only on O2-less System)

- 1.) Open an Unix Shell on the OC from the toolchest, become root by typing: `su root` ENTER.
- 2.) Type the root password and press ENTER. (The password is `#bigguy`)
- 3.) Type: `setdate mmddhhmmyyyy` (e.g.: `setdate 050409001999`) and press ENTER.

Note:

You may also type: `setdate` ENTER and you will be prompted through the individual entries.

Where:

`mm` is month (01-12)

`dd` is day (01-31)

`hh` is hour (00-23)

`mm` is minutes (00-59)

`yyyy` is year (1980-2030)

-
- 4.) The **setdate** program will set and then display the time and date for both the OC and the SBC. Verify the dates and times on the OC and the SBC are the same.
On an O2-less System, the time displayed for "SBC" is irrelevant, just ignore it.
-

Comment:

If the setdate script fails, refer to [Appendix F](#), page 69 for additional information.

On a System with an O2 Computer, if the dates or times are not the same, re-try [step 1](#) to [step 3](#) in this section, and re-verify the dates and times. If the dates or times are still not the same, change the Timezone to a "Universal" timezone as described in [Appendix C - "Changing The System Time Zone"](#), and then re-run this procedure.

- 5.) **exit** ENTER (to exit the root session).

2.17 Start up CT Applications

To start up the system, open a unix shell window and type: **startup** ENTER

2.18 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State MOD. If you don't have a System State MOD, skip this section (however, understand that you will have to perform ALL Characterizations and Calibrations in order to make the system operational).

Comment:

Image Works should not be displaying any images when you restore System State (to ensure preferences are restored correctly). If you were in Image Works earlier, you should go back and exit to the Browser.

- 1.) Insert the Save System State MOD into the MOD drive.
 - 2.) Click the SERVICE DESKTOP button.
-

Comment:

If the Service Key is installed, the user has to press the (1), (2), (3) keys in order to proceed.

- 3.) Press the PM option from the list.
- 4.) Press SYSTEM STATE.
- 5.) Click on ALL, which will highlight the cals, characterization, etc.
- 6.) Press RESTORE; the restore takes a few minutes.

Note:

If you have problems, see [Appendix F - "Error Messages And Troubleshooting"](#), beginning on page 69 for possible error messages and their recovery.

- 7.) Review the screen output for errors or missing files.

The scroll bar on the right works only when the tool is not busy performing some task. Please be patient, it takes a little while.

Note:

If you should get SAVE/RESTORE SYSTEM STATE ERRORS:

The Save/Restore System State process reports status on each file specification listed in the `/usr/g/config/SysState.cfg` file. The status may be:

- Save (or Restore) of {filename} succeeded
- Save (or Restore) of {filename} skipped. File not found
- FAILED

A status of `FAILED`, indicates potential problems, `skipped` means the file not found was not required. Please review the following for each file for which you see an error reported.

Some files simply may not exist on the system. If certain tools were never run then certain files may never get created. You may view `/usr/g/config/SysState.cfg` and look for a required field. `req=Y` means the file in question is required. You may also wish to check the system and MOD to verify whether it was actually missing or if there was a permission or owner problem with the file.

- 8.) Click NO for Reset Scan hardware pop up message.
- 9.) Click on DISMISS.

2.19 Downloading Flash Firmware

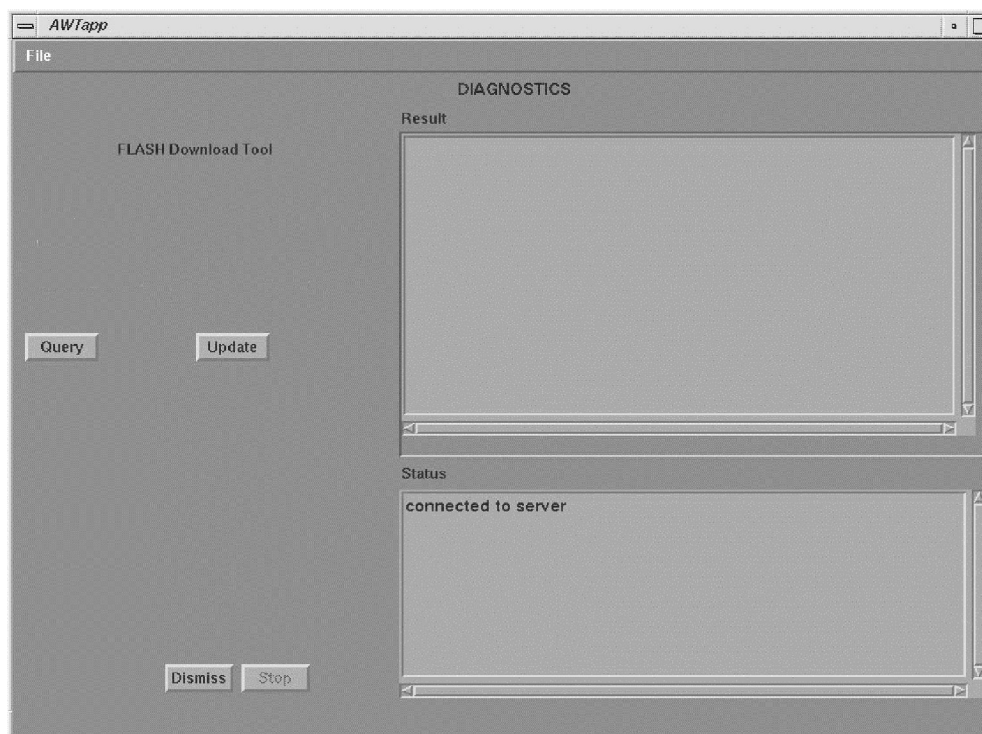


Figure 2-7 Flash Download Tool Graphical User Interface (GUI)

- 1.) If not already in the Service Desktop, click on SERVICE DESKTOP.
- 2.) Proceed to Flash Download Tool GUI:
 - a.) Select TOOLBOX/UTILITIES menu.
 - b.) Select INSTALL from the menu.

- c.) Select FLASH DOWNLOAD TOOL from the list.
- 3.) Click UPDATE button to start download.

Note: Ignore all pop-up messages by choosing CANCEL.

Comment: This will take a few minutes. Any invalid files should be re-downloaded by the software, and the output can be seen on screen.

- 4.) Click DISMISS to exit the graphical user interface.

2.20 Add Software Options

Software Options must be loaded every time the OC software is re-loaded. This program must be run once for each option MOD you are loading. For example, if you have two options MODs you must run this program twice.

- 1.) On Service Desktop, select TOOLBOX/UTILITIES -> INSTALL -> INSTALL OPTIONS. An Options Window appears.

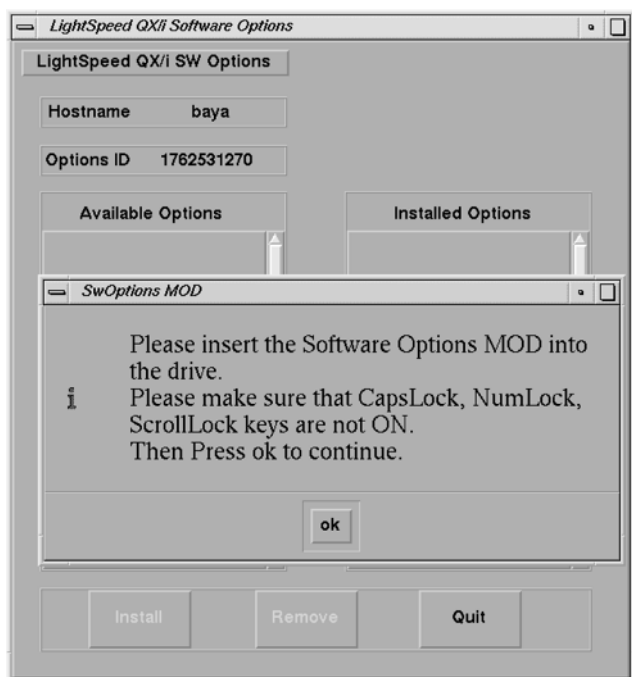


Figure 2-8 Options Window when First Selected

- 2.) Insert the Options MOD into the drive
- 3.) Click on OK

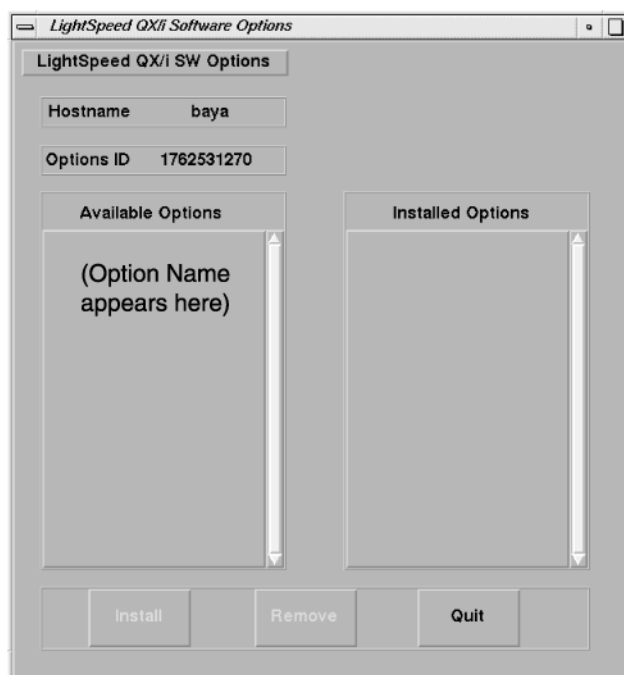


Figure 2-9 Example: Options Window

- 4.) Select the options to be installed. For example, click the left mouse button and drag to highlight all the available options and then release the mouse button.
- 5.) Click on the INSTALL button. A box may appear while the options are loading. When the option is displayed in the "Installed Options" list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half a minute (due to the fact that they are installing software).
- 6.) After the options are installed, select QUIT, then OK.

Comment: Do NOT reboot at this time, but continue as directed, and when you have completed installing all options, proceed to section [2.21](#).

- 7.) Remove the MOD & write protect the side with options.

Comment: The initial install requires the MOD to NOT be write protected; subsequent installs can be done with the MOD write protected.

- 8.) If you have another option MOD to load, go back to [step 1](#).

2.21 Install Cameras

Camera installation is required in the following cases:

- Loading a new system
- When a system state MOD is unavailable, unreadable or does not contain camera preference information
- If changes need to be made to existing camera configurations

To install cameras, proceed to [Appendix E - "Installing Laser & DICOM Cameras"](#).

2.22 Shutdown the System

- 1.) Press the pink SHUTDOWN button to bring down the system.
- 2.) Once the Applications and the OS have shutdown, press RESTART to bring the system back up again.

2.23 Verify System Operation

The following procedure makes sure the system is operating properly after upgrading or reloading the software.

- 1.) Open a UNIX shell and type: **hinv**
The output for the OC memory should be at least 512MB. If it is not, check the Host computer to make sure the memory is installed. If it is not installed, order the additional memory.

Note: The system must have at least 512MB Memory for proper operation. System performance will be seriously impaired if less than 512MB is present.

- 2.) ExamRx Desktop
 - a.) Start the system and make sure you are on the ExamRx desktop.
 - b.) Verify that the date and time shown in the system status area is correct.
 - c.) Perform Tube Warm-up and Fastcal.
 - d.) Set ExamRx Display up to Autoview using your favorite Autoview layout
 - e.) Scan your favorite image quality phantoms to ensure the system is making good images. Make more than one image in a series so you can play with the images below.
 - f.) Verify that the date and time shown on the images and in the browser are correct.
 - g.) Watch to make sure the images autoview.
 - h.) Use List/Select to bring an image up in an ExamRx free viewport.
 - i.) Try each of the Window/Level Preset keys to make sure they are set correctly.
 - j.) Set the window level via the mouse.
 - k.) Set the window level via the trackball.
 - l.) Use the Trackball to Page through the images.
 - m.) Film at least 1 sheet of film and verify the image quality.
 - n.) If the site has a Network Image Printer, verify the print quality.
- 3.) ImageWorks Desktop
 - a.) Go to the Image Works Desktop.
 - b.) Select Network and make sure all of the Remote hosts were restored when you restored the system state. If they were not restored, use the information you saved earlier (section 2.9 on page 20) to configure the site's networking.
 - c.) If you have any Genesis based systems configured for networking, and they are up and running, select one of the remote systems and **Query** to see if you can see those systems. *(If the Hospital and the user on a remote system grants you permission, try pushing images to the remote system.)* Use both genesis and non-genesis images if available.
 - d.) If you can see those systems, try pulling one or more exams onto this system.
 - e.) If it is all right with your site, try pushing images to all Genesis and DICOM systems that you have in your network configuration.
 - f.) Use Archive to restore one or more of the most recent Exams performed by the site.

- g.) Select an interesting Exam in the Browser and click on CT Viewer to see the images.
- h.) Try each of the Window/Level Presets to make sure they are set correctly.
- i.) Set the window level via the mouse.
- j.) Try a reformat of an interesting series.
- k.) If you have the 3D option, try doing a 3D rendering of an interesting series.
- l.) Print at least one film and check the image quality.
- m.) Remove one or more of the Exams you pulled onto the system.
- 4.) Service Desktop
 - a.) Go to the Service Desktop.
 - b.) Click on TOOLBOX/UTILITIES to see the list of Utility Applications.
 - c.) Click on TOOLS and then VERIFY SECURITY to see if the system can see your security key.
 - d.) Click on INSTALL and then VERIFY OPTIONS to see if the right options are configured.
 - e.) Click on SCAN ANALYSIS to see if you can see the scan files you made earlier.
 - f.) Click on SYSTEM RESETS, perform a SCAN HARDWARE RESET, and verify that it completes normally.
 - g.) When the Scan Hardware Reset is complete, click on CLEANUP to exit all the Service applications.

SOFTWARE LOAD IS NOW COMPLETE

2.24 Replace DASM In SCSI Chain

Replace the DASM in the SCSI chain by reversing the steps given in section 2.6 on page 18.

Appendix A

Procedure For Creating Default Partitions

PURPOSE

This Appendix contains the procedure for creating default partitions on the host (Octane) computer. Run this procedure only when a hard drive has been replaced, reformatted or is corrupted.

In systems that have an O2 Computer, the SBC (O2) computer does not need a procedure for manual creation of default partitions, because this partitioning is done automatically during the LFC.

In all systems, the Scan Data Disk is prepared by running reconfig, and selecting "Regenerate Database" (See "Appendix D" on page 55).

Note: Ensure the LightSpeed QX/i Operating System CD-ROM Disk #1 is in the drive before continuing.
If the disk has never been partitioned, an error condition is reported when starting to partition the disk: swap partition (1) is not a valid swap area .
This means you should continue the process to partition the drive.

Comment: If you need to format the Octane System Disk, see [Appendix F - "Error Messages And Troubleshooting"](#) page 72 for details.

PROCEDURE

- 1.) Select the ENTER COMMAND MONITOR, selection 5. A boot monitor screen will be displayed. Check to make sure the CD-ROM drive is NOT busy before you proceed. Start the standalone disk partitioning utility.

Type the command below.

- For the Octane Computer, the OC, type:

```
boot -f dksc(1,6,8)sash64 dksc(1,6,7)stand/fx.64 ENTER
```

- For the O2 computer, the SBC, disk partitioning is performed automatically.

- 2.) You will now be prompted for a response. Enter the responses that follow as indicated.

a.) Do you require extended mode with all options available(no)?**yes**

b.) fx: device-name = (dksc) ENTER

c.) fx: ctrl # = (0) ENTER

d.) fx: drive # = (1) ENTER

A message appears: SGI drive type = <Disk Model No>.

please choose one (? for help, .. to quit this menu) ---

[exi]t [d]ebug/ [l]abel/

[b]adbblock/ [ex]rcise/ [r]epartition/

e.) fx> r ENTER

--- please choose one (? for help, .. to quit this menu) ---

[ro]otdrive[o]ptiondrive[e]xpert

[u]srrootdrive[r]e[size]

f.) Select: [ro]otdrive

fx/repartition> ro ENTER

g.) fx/repartition/rootdrive: type of data partition = (xfs) ... ENTER

- h.) Continue? **y** ENTER
 - i.) A menu appears asking: Please Choose One.
 - j.) Go up a menu: **..** ENTER (i.e.: two periods)
 - k.) Type: **d/£i** ENTER(Note: There are no spaces in this command.)
 - l.) **fx/debug/fillbuf: buf offset = (0)** Just press ENTER
 - m.) **fx/debug/fillbuf: "fill string" = ()** Type: **a** and press ENTER
 - n.) **fx/debug/fillbuf: nbytes = (524288)** Type: **0x10000** and press ENTER
 - o.) A menu appears asking: Please Choose One.
 - p.) Type: **d/see** and press ENTER(Note: No spaces in this command.)
 - q.) **fx/debug/seek:blocknum = (0)** Type: **266240** and press ENTER
 - r.) A menu appears asking: Please Choose One.
 - s.) Type: **d/w** and press ENTER(Note: No spaces in this command.)
 - t.) **fx/debug/writebuf: buf offset = (0)** Just press ENTER
 - u.) **fx/debug/writebuf: nblocks = (4294967296)** Type: **128** and press ENTER
 - v.) A menu appears asking: Please Choose One.
 - w.) Type: **l/sy** and press ENTER (That is a lower case L, not a number one)(no spaces)
 - x.) Type: **exit** and press ENTER.
- 3.) You may now continue installation.

After performing disk partitioning on OC, See "Install System Software (IRIX 6.5 OS) on Operator Console (OC)" on page 25., to continue system software installation for the OC.

Appendix B

Reconfiguration Procedure

PURPOSE

This Reconfiguration Procedure describes how to modify the LightSpeed QX/i software configuration. This Reconfiguration allows most of the parameters entered during a LFC, to be changed without a complete LFC. The procedure consists of executing reconfig on the OC, changing the configuration on the screens, and accepting the reconfig (to allow the changes to be made).

Note: Reconfiguration only needs to be performed on the OC (Octane Computer). In systems with an O2 Computer, the SBC (O2 Computer) reconfiguration is performed automatically in parallel with the OC reconfiguration.

The following table indicates the parameter settings that can be changed using this Reconfiguration Procedure:

SYSTEM SETTINGS	PREFERENCES	HARDWARE SETTINGS	NETWORK SETTINGS
Hospital Name	Screen Display Language	Network Image Printer	Suite Name
Service ID	Preferred Fastcal kV		Host Name/AE Title
Time Zone	Date Display Format		Host IP Address
Next Exam #	Time Display Format		Net Mask
Scan Database Regeneration	Dose Information Display		Broadcast Address
			Default Gateway Address
			NIS (Yellow Pages)

Table 2-3 System Reconfiguration Settings

PROCEDURE

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window which closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

```
/var/adm/install.log.YYYMMDDWWHHMMSS
```

(Where *YYMMDDWWHHMMSS* is the Date/Time that the reconfiguration was started).

To view the file, type: `more /var/adm/install.log.YYYMMDDWWHHMMSS`

It is possible to abort the re-configuration while entering information on the re-configuration screens. Simply press the QUIT button at the top of the screen.

There is NO safe way to abort the re-configuration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the re-configuration, instead wait for it to complete, and rerun reconfig entering the correct parameters.

For systems containing an O2 computer, when you reconfigure the OC, the software automatically reconfigures the SBC as well.

- 1.) If Applications is running, you must shutdown system applications to the ctuser desktop.
 - a.) Click on the SERVICE DESKTOP button.
 - b.) On the desktop toolbar select TOOLBOX/UTILITIES.
 - c.) Select APPLICATIONS SHUTDOWN (to bring down applications only).
- 2.) On the OC, open a UNIX Shell window.
 - a.) Type: `su -` ENTER at the prompt.
 - b.) Enter the root (super user) password: `#bigguy`
- 3.) Change directory to scripts:
Type: `cd /usr/g/scripts` ENTER at the prompt.
- 4.) Launch the LightSpeed Install utility:
Type: `reconfig` ENTER at the prompt.

The OC displays the LightSpeed Install Utility Window as shown in [Figure 2-10](#).



Figure 2-10 Install Utility Window

- 5.) Using the mouse, click on the CONFIG button.

The OC displays the LightSpeed QX/i System Configuration - System Settings Screen as shown in [Figure 2-11](#).

Comment: The following pages show the screens which are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.

Figure 2-11 System Settings Screen

- 6.) Regenerate Database determines whether the Scan Database will be regenerated during reconfiguration.

Note: If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database. This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See [Appendix D - "Regenerating the Scan Database"](#) if you require more information regarding Scan Database Regeneration.

- 7.) Configure System Settings

- Hospital Name is the name that will show up on images produced by this scanner.
- Service ID is issued by the Service organization.

- c.) Select the Time Zone for the site.

Note: Use the scrollbar at the bottom of the time-zone selection list to view the entire description of the time-zone you are about to select, to ensure that you are selecting the correct time-zone for your location.

If the time-zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect. See [Appendix C - "Changing The System Time Zone"](#) if you require more information regarding time-zone setting & selection.

- d.) Next MOD # *Currently this field is not implemented.*
- e.) Next Patient Exam # configures the next Exam number the scan user interface will use. Get this information from the customer's patient log, or by examining the information contained in the file:
/usr/g/bin/scanRx.db (See "Record KEY System Information" on page 20.)
- f.) Next Diagnostic Exam # *Currently this field is not implemented.*
- 8.) Select the PREFERENCES button to display the Preference Settings Screen as shown in [Figure 2-12](#).

Figure 2-12 Preferences Setup Screen

Note: The Doctor's Title and Units for patient weight options are not currently implemented.

- 9.) Configure Preferences Settings
- a.) Language configures the language to be displayed on the Application screens (ENGLISH, FRENCH, GERMAN, ITALIAN).
- b.) Preferred Fast Cal KV configures the preferred kV that the Fast Cal Routine will

Comment:

calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 120 and 140.

These kVs should include all kVs which the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs

- c.) Date Format and Time Format configures the format in which the date and time will be displayed on the images.
- d.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose:
 - * Select ON (full CTDIw Display)
 - * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
 - * Select OFF (no CTDIw Display)

- 10.) Select the HARDWARE button to display the Hardware Settings Screen as shown in Figure 2-13.

Figure 2-13 Hardware Settings Screen

- 11.) Configure Hardware Settings
 - a.) DAS Type configures which DAS the software should expect. Presently, LightSpeed systems only support S DAS.
 - b.) Tube Type configures which X-Ray Tube the software should expect. Presently, LightSpeed systems only support MX_200MCT.
 - c.) PDU Type configures which PDU the software should expect. Presently, LightSpeed systems only support CT COMPACT.
 - d.) Collimator/Filter Type configures which Collimator/Filter the software should expect. Presently, QX/i systems only support HELIOS G1.
 - e.) Verify the presence or absence of an O2 computer in your console, and select YES or NO as appropriate. The default setting for this selection comes from a value maintained in the

system's INFO file.

Note: The O2 configuration selection can only be made during software load (not reconfig). If you have made the wrong selection during LFC, you must restart LFC from the beginning to correct this.

- f.) **Gantry Type** selects the type of Gantry connected to this console. Currently, QX/i systems only support H1 GANTRY for Gantry Type.
- g.) Select **Network Image Printer** only if the site plans to use a network connected postscript printer (such as Codonics printers) with this system. When selected:
 - * Enter the printer Hostname in the **Name** field.
 - * Enter the printer IP Address in the **IP Address** field.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e.: a computer with a printer attached); a Network Image Printer (which is supported) is a printer directly attached to the network.

- h.) **Camera Setup:**

Comment: Camera Setup is not performed using Reconfig, but rather is performed from the Service Desktop once Applications have been started. See [Appendix E](#) for details.

- 12.) Select the NETWORK button to display the Network Settings Screen as shown in [Figure 2-14](#).

Figure 2-14 Network Settings Screen

- 13.) **Configure Network Settings**

Comment: This screen provides the ability to declare the LightSpeed system on a hospital network. Key information such as Host Name, IP Address, Net Mask, and Default Gateway must be obtained from the hospital's network administrator.

a.) Enter the Suite Name.

Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters. Total must be four characters long. The name of the OC interface will be <Suite Name>_oc within the scanner's subnet, and the SBC interface will be <Suite Name>_sbc (for systems that contain an O2 Computer) within the scanner's subnet. Typically you should use CT01, unless the customer prefers a different Suite Name.

b.) Enter the hospital provided Host Name.

Note: The Host Name identifies the network hostname and AE Title of the LightSpeed system to the hospital's network.

The Host Name:

- * **MUST NOT** be <Suite Name>_oc or <SUITE NAME>_OC.
- * **MUST NOT** exceed 16 Characters.
- * **MUST** only contain the following characters: **A through Z, a through z, 0 through 9, - and _**

c.) Enter the hospital provided IP Address for the system.

d.) Enter the hospital provided Net Mask (if the LightSpeed system is on a subnet).

e.) Enter the Broadcast Address.

Comment:

The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used 0's.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

f.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.

g.) Select NIS (a.k.a. Yellow Pages database) Advanced Option only if requested by the hospital network administrator as follows:

- * Select ADVANCED OPTIONS button on the Network Settings Screen.
- * Select USE NIS button.
- * Enter the hospital provided Domain Name.

14.) Select ACCEPT on the LightSpeed QXi System Configuration Screen.

Comment:

The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC (and for systems containing an O2 computer, the SBC).

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window which closes when the loading process is complete. All the window output is logged to a file named:

/var/adm/install.log.YYYYMMDDWWHHMMSS

(Where YYYYMMDDWWHHMMSS is the Date/Time that the loading process was started).

- 15.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See [Figure 2-15](#).)



Figure 2-15 Reboot Screen

- 16.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.

APPLICATIONS START-UP

- 17.) In the Console shell, type: `st` ENTER

Appendix C

Changing The System Time Zone

Reconfig has the ability to change the system's time zone setting without having to perform a complete LFC.

This procedure will most likely be needed/used when the system is installed in a location where there is no local time zone selection available, or the time zone selected causes a failure when the `setdate` program is executed. (See "Set Date/Time on SBC & OC (or on OC Only if an O2-less System)" on page 35.)

- 1.) If Applications is running, shutdown Applications first by selecting TOOLBOX/UTILITIES -> SHUTDOWN APPLICATIONS from the SERVICE DESKTOP.
- 2.) Open the console window, by double-clicking on the console icon in the task box.
- 3.) In the console window, type `su -` and press ENTER
- 4.) Enter the root (super user) password: `#bigguy`
- 5.) Now type: `reconfig` ENTER
- 6.) Select CONFIG from the installation title screen
- 7.) On the System Configuration screen, select the Time zone for this site, and record it below:

Region / Time Zone

Table 2-4 Time Zone Setting

Note: If you are using this procedure because the `setdate` program failed to properly synchronize the OC & SBC times, then you should select the appropriate "Universal" time zone for your area of the world. In this case, automatic changes for Daylight Savings Time will not take effect.

- 8.) Press ACCEPT to accept re-configuration
- 9.) Now click YES to reboot the system.
- 10.) Proceed to Set Date/Time on SBC & OC (or on OC Only if an O2-less System) on page 35 to setup time.

Appendix D

Regenerating the Scan Database



NOTICE
Potential for
Data Loss

This procedure removes all scan and calibration data from the system disks. Make sure that all Images have been reconstructed before performing this procedure.

This procedure will re-initialize the Scan Data Disk, erasing all scan and calibration data. You will need to restore Calibrations (from System State), or re-calibrate the system once you have performed this procedure.

Reconfig has the ability to regenerate the Scan Database. The initial OC re-configuration screen (System Settings Screen on page 47) defaults the *Regenerate Database* selection to NO.

This procedure will most likely be needed/used when the Scan Data Disk has been replaced or reformatted. Using this procedure will eliminate the need to perform a complete LFC in these circumstances.

- 1.) If you need to regenerate the Scan Database, perform the Reconfiguration Procedure on page 45, and select the YES button for *Database Regeneration* on the System Settings Screen.
- 2.) Select ACCEPT, and allow the Re-configuration to complete.
- 3.) Once the Re-configuration has completed, and the system is rebooted, reload System Calibrations from the System State MOD, or if the State MOD is not available, perform all system Calibrations.

Appendix E

Installing Laser & DICOM Cameras



NOTICE
Potential For
Data Loss

Empty all filming queues before modifying camera parameters.

This Appendix contains procedures for recording important Camera setup information. Use the table(s) at the end of this section to record information from the set up screens. Use the directions in [step 9](#), page 64 to locate camera information.

If this is the initial installation of a DICOM Print Camera, see the System Installation Manual for additional, detailed information.

- 1.) Click on the SERVICE DESKTOP button.
- 2.) On the Desktop Toolbar select TOOLBOX/UTILITIES -> INSTALL -> INSTALL CAMERA. The LightSpeed QX/i Install Camera window appears, along with a warning message pop-up box. To remind you that all filming queues must be empty before you begin to update or delete a camera.
- 3.) The Graphical User Interface displayed shows a list of cameras installed ([See Figure 2-17](#)). First, you must click OK in the warning message box. [See Figure 2-16](#).

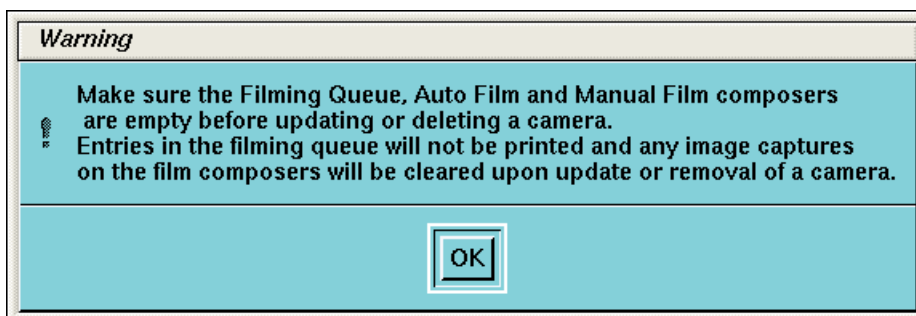


Figure 2-16 Warning Screen

- 4.) Now you are asked a series of question.
 - a.) To add a new camera, click the ADD button ([See Figure 2-17](#)).
 - b.) Now a dialog window for the camera type (DASM/DICOM) appears. If no DASM is detected during the OC boot, the DASM button will be disabled ([See Figure 2-18](#)). If a DASM is present and has not been detected, re-boot the OC and run the camera configuration tool again.

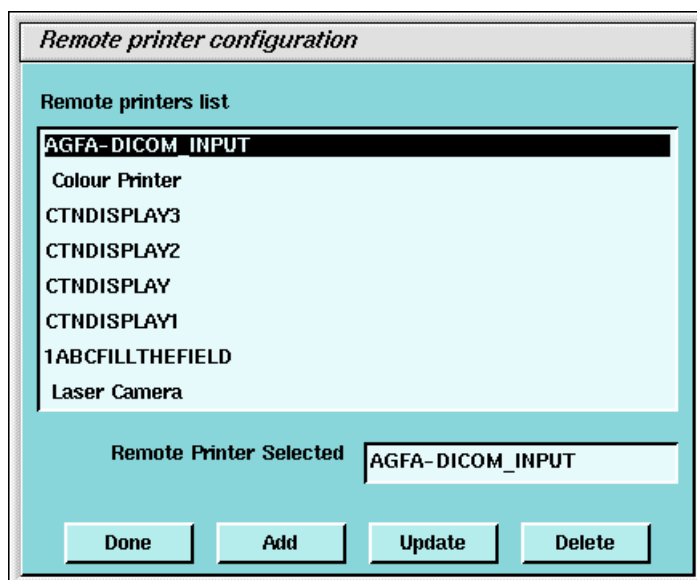


Figure 2-17 Printer Configuration GUI

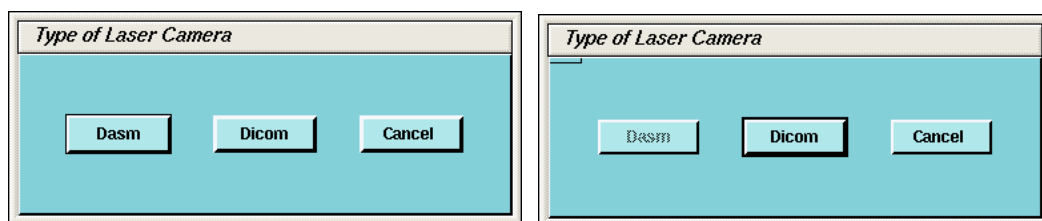


Figure 2-18 Dialog Box for camera Type (with and without DASM)

- 5.) To add a new laser camera, click DASM in the camera type dialog box. This brings up a list of available camera models. Select the appropriate model from the list and click SELECT (See [Figure 2-19](#)). Now you must configure it.

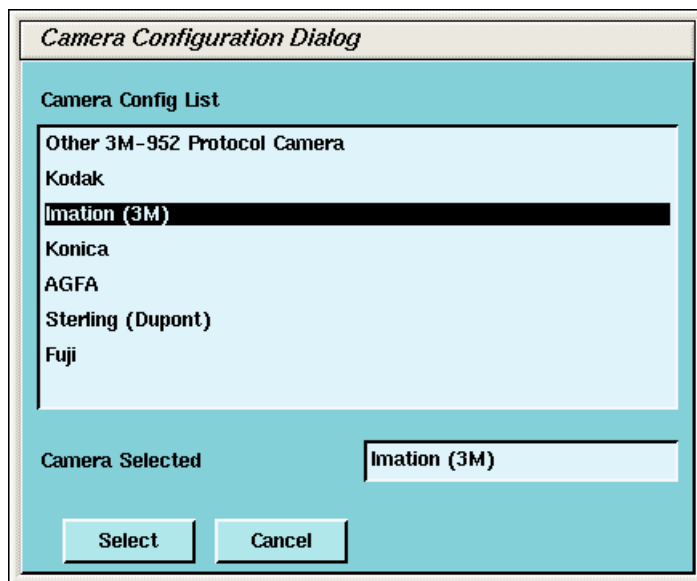


Figure 2-19 Camera Model Dialog (with DASM)

- a.) DASM Interface is automatically detected as either Analogue or Digital
- b.) Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.
- c.) Camera manufactures provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

Comment:

To film good images, and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak: SMOOTH
 Dupont/Sterling: SMOOTH
 3M/Imation (Laser Camera): SHARP
 3M/Imation (Dry View): SMOOTH
 Agfa: SMOOTH

- d.) Select the appropriate File Format. Select ON from the drop down list boxes on the menu. Valid film formats are determined by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4). The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4). When finished setting parameters, click on DONE and proceed to step 8.

Laser Camera Configuration

DASM Interface : Analogue

Laser Options : ☒ Slide
☒ Zoom

Magnification Type : ☒ Smooth
☐ Sharp

Film Format

F_1x1 <u>On</u>	F_2x1 <u>On</u>	F_2x2 <u>On</u>	F_3x2 <u>On</u>
F_3x3 <u>On</u>	F_4x2 <u>On</u>	F_4x3 <u>On</u>	F_4x4 <u>On</u>
F_5x3 <u>On</u>	F_5x4 <u>On</u>	F_6x4 <u>On</u>	

Done Cancel

Figure 2-20 Laser Camera Configuration

- 6.) To add a new DICOM camera, click on ADD and then DICOM in the dialog box that appears.

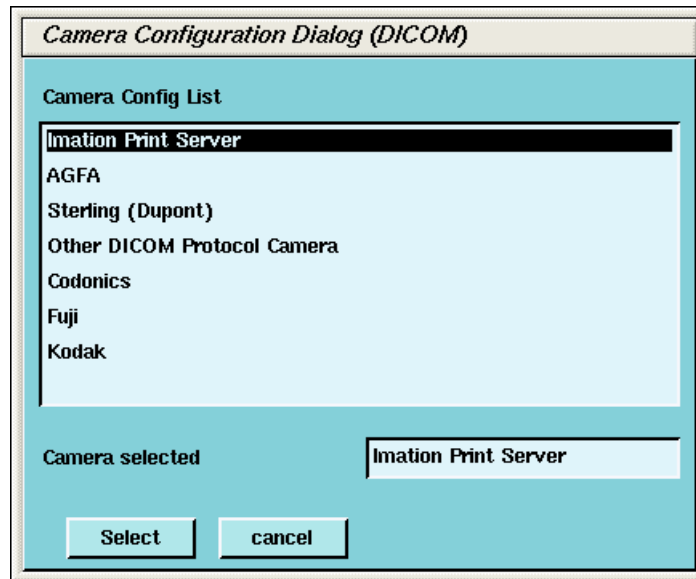


Figure 2-21 Camera Model Dialog (DICOM)

- a.) A list of camera models appears ([See Figure 2-21](#)). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that models except the Network parameters.

Selection of a different camera model clears the Image Quality parameters, because these are camera manufacture dependent.

- b.) Enter the Network Parameters ([See Figure 2-22](#))

- > Device Name A unique name used to identify the camera.
- > Host Name DICOM print server host name, as defined by the hospital.
- > IP Address DICOM print server IP address, as defined by the hospital.
- > AE Title DICOM print server application entity title, as defined by the print server. *You should consult the manufacturers DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site specific. Make sure that you check with the Camera Manufacturer's Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination DICOM Print Camera.

- > TCP/IP Listen Port DICOM print server TCP/IP listen port, as defined by the server. *You should consult the manufacturers DICOM Conformance Statement.*
- > Comments Optional comments used by the DICOM print server.

*Comment:
It's advised to
re-check the
preset
information with
the camera
manufacturer's
representative.*

Figure 2-22 DICOM Camera Configuration

- c.) Medium Type specifies the type of film being used. Currently, only BLUE FILM and CLEAR FILM are supported.
- d.) Set Destination to the final location for film output. This is either MAGAZINE or PROCESSOR.
- e.) Orientation selects film orientation. Only PORTRAIT is currently supported.
- f.) Set the Magnification Type. This parameter selects the algorithm used to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacture to ensure optimal image quality. Choices are describe below:

- > None No interpolation. This option is not supported by all cameras.
- > Replicate Adjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normal preferred.*
- > Bilinear A 1st order interpolation of pixels. Results in images usually described as blurred. *This algorithm is not normal preferred.*
- > Cubic A 3rd order interpolation. Used with a large number of possible formulations. Camera manufactures define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing soefficients" is camera dependent.

Comment: For most Camera Manufacturers, the preferred selection is CUBIC.

- g.) Select the Standard Film Formats. Select the film format by choosing ON in the pull-

down menu box located below each selection. See Figure 2-22. Valid film formats are set by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).

- h.) After the camera has been configured, click the ADVANCED button. This creates the camera device file for you automatically and pops up the Advanced Parameters screen. See Figure 2-23.

Figure 2-23 Advanced Parameters (Camera Attribute Dialog)

- 7.) Advanced camera parameters control the image quality of films.

Note: For more information on the proper settings for these parameters, please see the Camera's DICOM Print Device Conformance Statement or the Camera Manufacturer Representative. The detailed DICOM Conformance Statement for LightSpeed QX/i is available as Direction 2211216-100. You may need to refer to a copy of this document as you are working with the Camera Manufacturer's Representative, to correctly set up the DICOM Print Camera I/Q and Time-out Settings.

- a.) **Configuration** - This parameter is camera manufacturer dependent as is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field will scroll automatically as text is entered. To review your entry, simply click and hold the middle mouse button, while the cursor is in the field, and drag the mouse towards the right (or left) as needed.

Note: **Recommended Configuration Setting Values:**

Agfa Drystar (MG3000)	PERCEPTION_LUT=KANAMORI (100)
Imation Dryview (8700)	LUT=0, 7
Kodak Laser Printer 190	CS434\CN0\PD1.20

- b.) **Smoothing Type** - Set Smoothing Type to ON, and input the selected value. This

parameter is used when the magnification type is CUBIC. It represents the coefficient for the image resolution algorithm. This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.

Note:

Recommended Smoothing Type Starting Values and Ranges:

Agfa DryStar (MG3000)	Start Value:140	Range: 137 - 150
Imation Dryview (8700)	Start Value:3	Range: 3 - 13
Kodak Laser Printer 190	Start Value:Enhanced	Range: Normal

- c.) **Minimum and Maximum Density** - Used to set brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacture dependent. For **Maximum Density**, input the correct value into the text box. For **Minimum Density**, set it to ON and input the correct value in the text box.

Note:

Recommended Minimum and Maximum Density Starting Values:

Agfa Drystar (MG3000)	Min.: 20 or 23	Max: 300
Imation Dryview (8700)	Min.: (Blank)	Max: 300
Kodak Laser Printer 190	Min.: 20	Max: 300

- d.) **Empty Image Density** - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- e.) **Border Density** - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- f.) **Film Size** - Allows the system to specify a particular film size, if selected.
- g.) **Trim** - YES produces a white (clear) box surrounding each image.
- h.) **Priority** - This sets the print priority.
- i.) If you have completed entry of advanced parameters, click DONE.
- 8.) Go to [section 2.22 on page 40](#) for continuing installation if you are performing a LFC.

- 9.) **Camera Data Tables:** To locate Install Camera Information: Click on the SERVICE DESKTOP button. On the Desktop Toolbar select TOOLBOX/UTILITIES -> INSTALL -> INSTALL CAMERA. The LightSpeed QXi Install Camera window appears. Select each of the cameras that are installed from the list of installed cameras, and click on UPDATE to view the camera's settings. Record the values used to setup each camera in the tables that follow. Extra tables are provided for multiple cameras

Note: You can determine this information by looking at the contents of the following files:

- For a DASM Camera: `/usr/g/ctuser/app-defaults/devices/camera.dev`
- For a DICOM Print Camera: `/usr/g/ctuser/app-defaults/devices/name.cfg` where, `name.cfg` is the camera device name from the printer configuration GUI.

Example: `more <filename from above> ENTER`

DASM CAMERA #1

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	☐ Digital ☐ Analog
Options	Slides or Zoom	☐ Slides ☐ Analog
Film	Smooth or Sharp	☐ Smooth ☐ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 2-5 DASM Camera #1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 2-6 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 2-7 DICOM Camera #2

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 2-8 DICOM Camera #3

Appendix F

Error Messages And Troubleshooting

Tables of potential error messages and potential solutions are provide on the pages that follow.

Section	Error	Possible Cause	Suggested Recovery Solution
Install System Software (IRIX 6.5 OS) on Operator Console (OC) on page 25, step 4	dks1d6s8: drive not ready	CD-ROM was busy when the machine tried to mount it.	Exit menu and try again. Type: <u>ENTER</u> Continue from beginning of Section 2.13
Install System Software (IRIX 6.5 OS) on Operator Console (OC) on page 25 step 6 and (sub) step b	OS Hangs. Does not accept user input	IRIX6.5 problem	Restart system and restart installation at Section 2.11
Installing System Software (IRIX 6.5 OS) on the SBC (for Systems with an O2 Computer) on page 28	No matches found	System unable to locate entered filename	Check command for misspellings and reenter.
Installing System Software (IRIX 6.5 OS) on the SBC (for Systems with an O2 Computer) on page 28	Host oc not responding, retrying or Percentage complete suddenly stops increasing.	Connection between OC and SBC failed, usually due to the missing route.	Attempt to establish a default gateway. <i>Open a new UNIX window, type: route delete default</i> Type: route add default 192.9.220.1 If progress does not continue, check cables and restart procedure from beginning of Section 2.13 .
Installing System Software (IRIX 6.5 OS) on the SBC (for Systems with an O2 Computer) on page 28	Specific Errors unknown.	Helios_SRU_Install package not installed. (Command 2.8.18 skipped) To verify package installation, type: <u>showprods</u> <u>ENTER</u> on the SBC. If the Helios_SRU_Install package is not present, it was not installed	The Helios_SRU_Install package must be installed and then the procedure can be restarted. On the SBC, as root type: ifconfig ec0 192.9.220.1 up <u>ENTER</u> On the OC, as root type: route add default 192.9.220.1 <u>ENTER</u> exportfs -ua <u>ENTER</u> exportfs /CDROM <u>ENTER</u> On the SBC, type: mkdir /CDROM <u>ENTER</u> mount 192.9.220.10:/CDROM /CDROM <u>ENTER</u> cd /CDROM/dist <u>ENTER</u> inst -A -f ./Helios_SRU_Install <u>ENTER</u> cd / <u>ENTER</u> umount /CDROM <u>ENTER</u> /sbin/configNetst.sru -snip 192.9.220 <u>ENTER</u> On the OC, type: eject 1 6 <u>ENTER</u> Resume the LFC procedure at Section 2.14 , page 28.

Table 2-9 Potential LFC Error Messages and Recovery Solutions

Section	Error	Possible Cause	Suggested Recovery Solution
Install QX/i Software on the OC and SBC (or on the OC Only, on an O2-less System) on page 30	Specific Errors unknown.		Review the installation log file: <code>more /var/adm/install1.log.YYYYMMDDWWHHMMSS</code> (where YYYYMMDDWWHHMMSS is the date/time that the software installation was started. Restart the Applications Software Installation at Section 2.15 , page 30.
Set Date/Time on SBC & OC (or on OC Only if an O2-less System) on page 35 step 3	<code>setdate</code> command failed to set times (only on Console with O2 Computer).	SBC may not be up. To verify, type: <code>ping sbc</code> <u>ENTER</u> The screen should show a new line of text every second indicating packets being returned if the SBC is up. If new text does not appear, <u>CTRL-C</u> out and continue with the box on the right.	From a UNIX window on the OC, <code>cu</code> to the SBC and check if it is up. In a UNIX shell window, type: <code>cu sbc</code> <u>ENTER</u> If the system responds with: <code>System starting up</code> Wait until the systems comes up to a prompt and retry the <code>setdate</code> command in an OC window. If the SBC does not appear to be responding, exit the “cu” session, halt the OC, cycle power to the console, and try again.
Set Date/Time on SBC & OC (or on OC Only if an O2-less System) on page 35 step 4	Times do not match between OC and SBC (only on Console with O2 Computer).	Failed <code>setdate</code> command	Retry the <code>setdate</code> command.
Set Date/Time on SBC & OC (or on OC Only if an O2-less System) on page 35 step 4	Time zones do not match between SBC and OC (only on Console with O2 Computer)	Time zones do not match	Execute procedure in the next line of this procedure to fix the time zones.
Set Date/Time on SBC & OC (or on OC Only if an O2-less System) on page 35 step 4	Time zones do not match between SBC and OC (only on Console with O2 Computer)	Previous deviations from standard load, failure of <code>setdate</code> command.	FTP the correct file from the OC to the SBC. Open a UNIX shell on the SBC and type: <code>rcp oc:/etc/TIMEZONE /etc/TIMEZONE</code> <u>ENTER</u> Or, perform Changing The System Time Zone on page 53.
Restore System State on page 36 step 6	Corruption detected on the MOD in the drive. MOD cannot be mounted. MOD initialization completed with failure Save/Restore System State: Completed with failures.	Bad or corrupted MOD.	Attempt to repair MOD with <code>fsck</code> and reissue command. Open a new UNIX window, insert MOD in drive and type (as root): <code>fsck -n /dev/dsk/dks1d3s7</code> <u>ENTER</u> If <code>fsck</code> fails or system hangs, the system state cannot be loaded from the MOD and the system must be re-calibrated and manually re-configured. Use re-configuration data recorded in Section 2.9 If <code>fsck</code> completes successfully, restart procedure at Section 2.18

Table 2-9 Potential LFC Error Messages and Recovery Solutions (Continued)

Section	Error	Possible Cause	Suggested Recovery Solution
Appendix A	FORMATTING PROCEDURE		Issue commands in Appendix A up to and including step 2, step d. For the next step, select [f]ormat by typing: f <u>ENTER</u> fx/format: Drive parameters to use in formatting = (current) Type: <u>ENTER</u> format will take approximately 90 minutes . . This is seldom necessary and may cause drive problems * * * * * WARNING * * * * * about to destroy data on disk dksc(0,1,0)! Ok? Type: y <u>ENTER</u>

Table 2-9 Potential LFC Error Messages and Recovery Solutions (Continued)



GE MEDICAL SYSTEMS

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